PHENIX™ SYSTEMS

Additive manufacturing systems using laser sintering

All about Phenix Systems

Innovation, development and production times, the search for energy, material and cost savings, quality and flexibility constitute key elements in manufacturing processes. Phenix Systems is actively involved in this technological evolution and provides solutions and systems using a patented laser sintering technology.

This revolutionary rapid process for manufacturing industrial and technical parts or series of parts is carried out without any tooling by using a 3D digital file, and consists in soldifying - layer by layer - particles of metallic or ceramic powders with a laser beam.

PHENIX SYSTEMS, a whole length ahead

PHENIX SYSTEMS was born in 2000 from the conviction that in the future rapid layer manufacturing equipments would become a standard and open the way to new possibilites.

Possessing a patented technology the company is positioned as a specialist - which constitutes a very strong competitive advantage. Phenix Systems has the only solution on the market that offers the possibility of using metallic and ceramic powders. Thanks to this know-how the company is one of the world leaders for laser sintering production equipment.

A tried and tested ground-breaking technology

Laser sintering rapid manufacturing is entirely different from the traditional techniques that consist of producing an object from raw material (machining, spark-erosion, micro-casting, laser cutting,...).
As a mature high technology, laser sintering replaces these various techniques and is rapidly becoming established in various sectors and in very wide fields of application: the motor industry, aero-space, energy, luxury, bio-medical,....

The major advances benefiting users are:

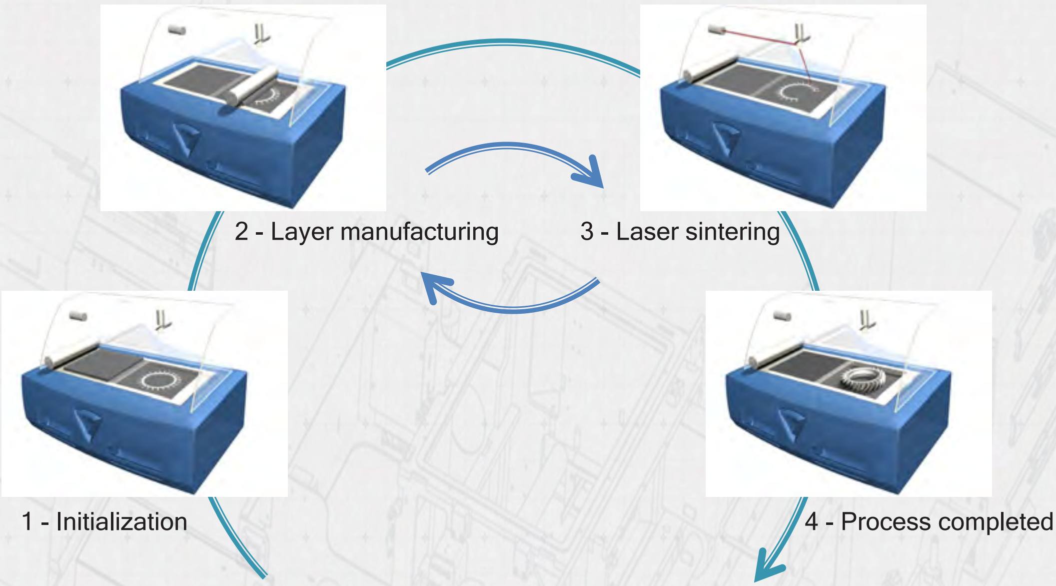
- *The acceleration in the cycle of development,*
- *The increases in productivity, material and energy savings,*
- *The use of new alloys,*
- *The access to innovative ideas.*

Discover Phenix Systems' offer

Phenix Systems - a global expertise

Process

The parts are directly created from their three-dimensional perception in CAD software. Each representative section of the part is laser sintered in solid phase with or without instantaneous consolidation at the melting point depending on the materials being used.

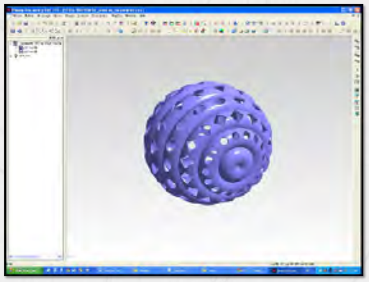


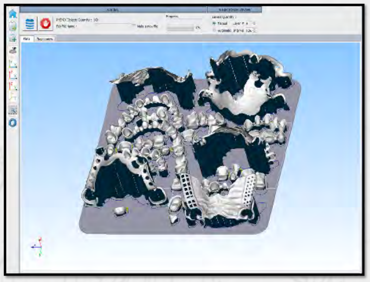
Design and integration of the machines

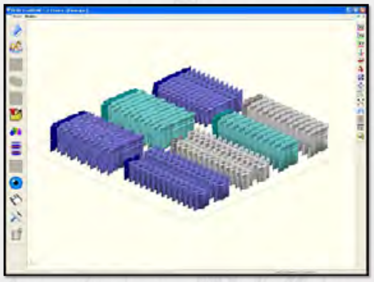
The systems are completely designed and assembled on Phenix Systems premises based at Riom (France). Every model of equipment has its own line of integration.

CAD/CAM software development

Phenix Systems develops its own CAD/CAM software: Phenix Processing, Phenix Manufacturing, Phenix Dental,.... An internal research department is constantly working on improving these tools that are essential for manufacturing parts. These programs are designed and optimised for the systems.

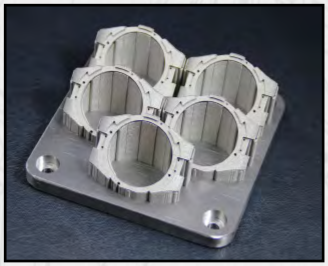






Customer guidance

Phenix Systems advises its customers in their product development phases so that they fully benefit from all process's advantages. On request, producing tests and pre-series qualification parts is also possible. This offer enables the companies that are interested to verify that the technology is well adapted to their needs.



Marketing the products

Thanks to its innovative solutions, Phenix Systems Group counts customers all over the world. For the North American market, the technico-commercial centre is the company AMT Inc. - a 100% Phenix Systems subsidiary.

Powders conception

The company SINT-TECH, ISO 9001 and ISO 13485 certified, proposes powders suitable for the Phenix process depending on the applications.

Systems

The range of «PX» systems developed these last three years, integrates, depending on the type of application, all the functions that enable a combination of parameters to be adapated so as to use metallic and ceramic powders with perfectly controlled grading.

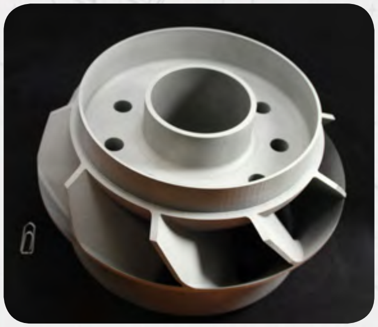
Particular attention has been paid to the ergonomics of the equipments. The powders are handled in complete safety. The range is available in three models, each adapted to the most varied technico-economic issues.

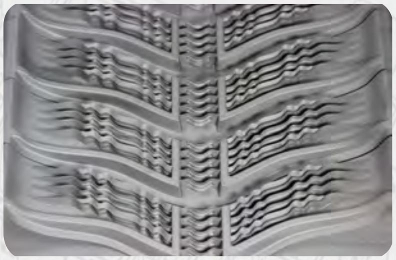
PXS

PXM

PXL

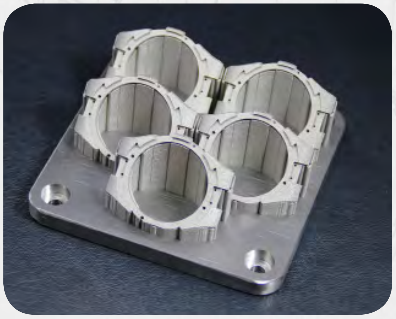
PXL SYSTEM

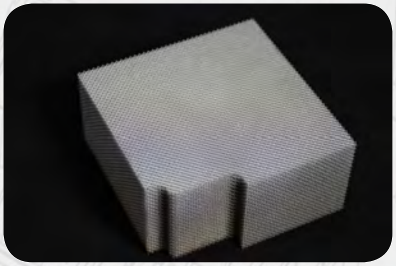




TECHNICAL DATA	
Fiber laser	P=500 W - λ=1070 nm
Layering system	Adjustable
Building volume	250 x 250 x 300 mm
Metal materials	Stainless steels, tool steels, non-ferrous alloys, super alloys,...
Ceramic materials	Alumina, cermet,...
Repeatability	x=20 µm ; y=20 µm ; z=20 µm
Detail size	x=100 µm ; y=100 µm ; z=20 µm
Loading system	Automatic
Recycling system	Automatic
CAD/CAM software	Phenix Processing - Phenix Manufacturing
Control software	PX Control
CAD read formats	IGES, STEP, STL
Dimensions	L= 2,40 m; W= 2,20 m; H= 2,40 m
Weight	About 5000 kg
Energy (power supply requirements)	Max. 15 KVA - 400 V Tri
Pneumatic	6-8 bars
Certification	CE
Class 1 laser machine in standard production. CEI 60825-1 (2007) NF EN ISO 11553-1 (2009)	

PXM SYSTEM

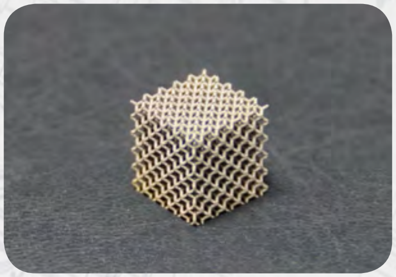


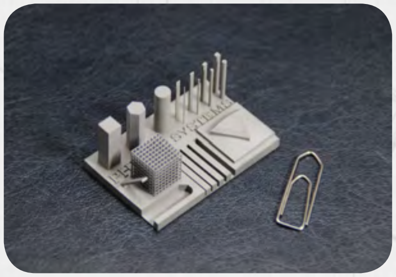


TECHNICAL DATA	
Fiber laser	P=300 W - λ=1070 nm
Layering system	Adjustable
Building volume	140 x 140 x 100 mm
Metal materials	Stainless steels, tool steels, non-ferrous alloys, super alloys,...
Ceramic materials	Alumina, cermet,...
Repeatability	x=20 μm ; y=20 μm ; z=20 μm
Detail size	x=100 μm ; y=100 μm ; z=20 μm
Loading system	Semi-automatic
Recycling system	Optional external system (PX BOX)
CAD/CAM software	Phenix Processing - Phenix Manufacturing*
Control software	PX Control
CAD read formats	IGES, STEP, STL
Dimensions	L= 1,20 m; W= 1,50 m; H= 1,95 m
Weight	About 1500 kg
Energy (power supply requirements)	Max. 10 KVA - 400 V Tri
Pneumatic	6-8 bars
Certification	CE
Class 1 laser machine in standard production. CEI 60825-1 (2007) NF EN ISO 11553-1 (2009)	

* *The PXM system is available in Dental version thanks to the Phenix Dental CAD/CAM software.*

PXS SYSTEM





TECHNICAL DATA	
Fiber laser	P=50 W - λ=1070 nm
Layering system	Adjustable
Building volume	100 x 100 x 80 mm
Metal materials	Stainless steels, tool steels, non-ferrous alloys, super alloys,...
Ceramic materials	Alumina, cermet,...
Repeatability	x=20 µm ; y=20 µm ; z=20 µm
Detail size	x=100 µm ; y=100 µm ; z=20 µm
Loading system	Manual
Recycling system	Optional external system (PX BOX)
CAD/CAM software	Phenix Processing - Phenix Manufacturing*
Control software	PX Control
CAD read formats	IGES, STEP, STL
Dimensions	L= 1,20 m; W= 0,77 m; H= 1,95 m
Weight	About 1000 kg
Energy (power supply requirements)	Max. 3 KVA - 230 V Mono
Pneumatic	6-8 bars
Certification	CE
Class 1 laser machine in standard production. CEI 60825-1 (2007) NF EN ISO 11553-1 (2009)	

** The PXS system is available in Dental version thanks to the Phenix Dental CAD/CAM software.*

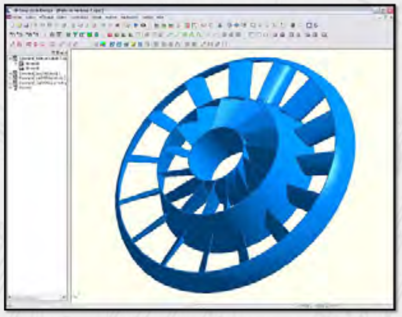
CAD/CAM software

Phenix Processing

PHENIX Processing is a CAD/CAM tool that integrates all the technical data involved in the laser sintering process. This application enables you to define all your manufacturing parameters user-friendly and precisely. This software is available on the three systems.

THE CAD FUNCTIONS

- Solid modelling
- Surface modelling
- Mesh creation
- Accepted file formats: STEP, IGES, STL



THE CAM FUNCTIONS

- Programming of laser parameters
- Positionning objects in the manufacturing volume
- Programming of layering parameters
- Calculation and visualisation of the layer by layer laser trajectories
- Creation of the manufacturing files

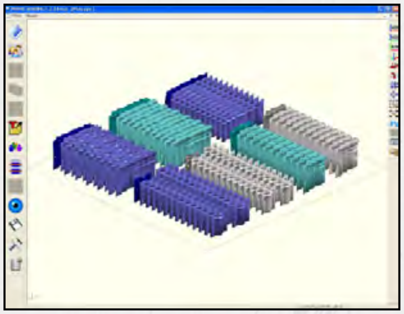
Advantage: open and intuitive software leaving the user a wide liberty

Phenix Manufacturing

PHENIX Manufacturing is a simple and intuitive solution specifically dedicated to the manufacturing of parts by type of application. The user is guided step by step from the importation of the CAD elements to the creation of the manufacturing files. This software is available on the three systems.

THE FUNCTIONS

- Managing of the parts typologies
- Positionning 3D models on the manufacturing plate
- Calculation and visualisation of layer by layer laser trajectories
- Creation of the manufacturing files



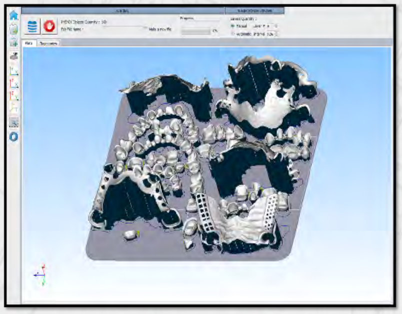
Advantage: the software evolves to suit the user's specific needs

Phenix Dental

PHENIX Dental is a high performance solution for programming the manufacturing of fixed and removable prosthesis. The user is guided step by step from the importation of dental files to the creation of the manufacturing files.

THE FUNCTIONS

- Optimised positionning 3D models on the manufacturing plate
- Automatic generation of production supports
- Calculation and visualisation of layer by layer laser trajectories
- Anti-collision detection
- Automatic creation of manufacturing files



Advantage: user interface designed to be pleasant and ergonomic

Materials

In the laser sintering process controlling the layer of the pulverulent material is fundamental. A programmable and parametrable system patented by Phenix Systems makes it possible to adapt the form and average size of the particles of metallic and ceramic powders.

A wide range of materials varieties is available:

Metal : Stainless steel, tool steel, non-ferrous alloys, super alloys, precious metals,...

Ceramic : Alumina, cermet,...

The company SINT-TECH, ISO 9001 and ISO 13485 certified, proposes a range of powders suitable for the process developed by Phenix Systems.

These powder’s gradings have been selected to guarantee an optimised result for their implementation with the systems of the «PX» range and the former «PM» range.

An innovative offer

Technicality and quality

- Possibility of producing parts with complex geometry,
- Patented system for layering fine graded powders,
- Reproductibility and process stability.

Flexibility and reactivity

- Reduced design cycle and rapid prototyping during product development,
- Easy to implement.

Cost and productivity

- Only the material used is consumed,
- No manufacturing tools needed,
- Technology perfectly adapted to mass production.

Services

- Assisting customers during the product's development phase,
- Research and development for new materials,
- Adapting CAD/CAM programs to customers' specifications.

More information:

www.phenix-systems.com

Do you want to know more about PHENIX SYSTEMS? We invite you to come and discover our website. There you can consult our whole range and download technical files. It also gives you the chance to keep up to date with all the latest news about the Phenix Systems Group in Europe and all around the world: fairs, technological innovations, conferences, etc....
All our world-wide distributors are listed under the heading «contact» according to your area, do not hesitate to contact them.

Please scan this QR-code with your smartphone to visit our website.

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